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The development of the student online learning patience scale (SOLPS)

Ali Eryilmaz[1] · **Ahmet Basal**[2]

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Abstract
Learning does not happen immediately; it takes time and effort. Thus, patience is a virtue required for any type of successful learning. Given the increasing prevalence of online learning, understanding students' patience is necessary to devise ways to sustain it. In this study, we define online learning patience as the ability to steadily continue studying or practicing in the online learning environments, even when it becomes difficult or takes a long time. The present study aimed to develop and validate a new instrument, the Student Online Learning Patience Scale (SOLPS), using a cross-sectional research design. The study participants, distributed across four distinct groups: Exploratory ($n=250$ students), Confirmatory ($n=266$ students), Validity for Impulsivity and Test-anxiety ($n=486$ students), and Validity for Self-control ($n=232$ students). The items in the scale were based on the literature and analysis of interviews with students. The scale structure was examined through Exploratory Factor Analysis (EFA) and Confirmatory Factor Analysis (CFA), and reliability and validity were tested. The final SOLPS consists of 10 items with one dimension and can be used to create teaching methods and strategies that increase students' online learning patience.

Keywords Learning · Online learning · Online instructional environments · Patience · Scale development

✉ Ali Eryilmaz
erali76@hotmail.com

[1] Department of Psychological Counselling and Guidance, Yildiz Technical University, İstanbul, Turkey

[2] Department of Educational Sciences, Yildiz Technical University, İstanbul, Turkey

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1 Introduction

In online environment, learning has been transferred from physical classrooms to online platforms with distance separating students and teachers, raising various problems for both. Apart from the technical and organizational challenges of this learning environment, student-related problems emerged that interfered with their learning, such as their lack of active participation and engagement (Downes, 2016; Foster et al., 2018), increased responsibility for their learning (Henderikx et al., 2019), lack of motivation (Almanthari et al., 2020; Chen & Jang, 2010; Henderikx et al., 2019; Muilenburg & Berge, 2005), distractions from the physical environment (Barratt & Duran, 2021; Schmidt, 2020), divided attention due to increased multitasking behaviors during online instruction (Lepp et al., 2019), boredom (Chang, 2020), and socializing (Muilenburg & Berge, 2005). These challenges significantly affect the online learning experience, often leading to reduced effectiveness in education and hindering the overall academic progress of students. In online learning, the learners, we believe, should demonstrate learning patience to overcome these challenges inherent in this instructional environment.

Online learning patience is particularly crucial in addressing the issue of physical separation between students and teachers, a characteristic feature of online learning. With online learning patience, students can better handle the increased responsibility for their own learning, a significant shift from traditional classroom settings. It also aids in reducing the impact of distractions from the physical environment and the tendency towards multitasking, which can undermine the learning process. Furthermore, online learning patience is instrumental in combating boredom, a common challenge in online environments where the usual dynamics of in-person interaction are absent. By embracing online learning patience, learners can maintain a steady and focused approach to their studies, ensuring a more productive and fulfilling online educational experience. However, there is no scale that determines whether students have online learning patience. Considering the apparent gap in research regarding patience within online learning environment, there lacks a thorough, domain-specific exploration of patience, indicating a compelling need for developing and validating an online learning patience scale, the aim of the current study.

1.1 Exploring patience

In the literature, patience has been dealt with as a general personal trait rather than a specific attribute that can be measured. The purpose of developing a scale for online learning patience is to locate patience as a field-specific trait for the online learning environment. To this end, we first start with studies in which patience is treated in a general manner. Patience has been identified as one of the most important human characteristics (Stevens et al., 2005), and according to evolutionary psychologists, it is one of the important traits that has helped the species survive (McClure et al., 2004). Patience is generally associated with staying calm, maintaining composure, enduring negative experiences such as waiting without complaining and continuing action with determination (Hanks & Stratton, 2007; Morinis, 2007). It is also seen as a trait that can be developed (Alan & Ertac, 2018). Schnitker and Emmons (2007)

identified behavioral (e.g., calm waiting) and emotional (e.g., little negativity) as indicators of patience and found an association between high levels of patience and both physical and mental health. In the psychological dimension, Schnitker (2012) earlier proposed three domains of patience: interpersonal (patience with others), daily life (patience with ordinary tasks and impediments), and hardship (patience with major challenges and setbacks). Impatience, therefore, is defined in terms of opposites of these traits, such as difficulty maintaining composure when kept waiting or delaying gratification, and it is considered a characteristic of a Type A (achievement-oriented) Personality (Spence et al., 1987).

It is generally agreed that people should cultivate patience for both social and personal reasons. Studies have shown that patient individuals are more cooperative (Curry et al., 2008; Stevens & Hauser, 2004). Patience is seen as necessary for the healthy functioning of a family, effective in children's development of healthy habits of consumption and money management, (Bettinger & Slonim, 2007), and important in character development, whereas impatience is associated with child abuse and neglect (Hanks & Stratton, 2007). On the personal level, patient individuals cope better with negative situations. Khormaei et al. (2014) concluded that as individuals' patience levels increase, their hope levels also increase. Patience, particularly in the form of calm waiting and delay of gratification, is also considered a virtue in most religions (McCullough & Carter, 2011). In summary, the existing literature predominantly views patience as a broad personal characteristic rather than as a specific, measurable attribute relevant to particular fields like online learning.

1.2 Conceptual clarifications: exploring related concepts with patience

The concept of patience is associated with many other concepts. At the forefront of these are self-control, resilience, engagement, and persistence. While these concepts are related to patience, they are also distinct from the concept of patience. Each of these terms, although associated with patience, carries its unique nuances and meanings that differentiate it from patience. In the following lines, we aim to elucidate the intricacies of patience and its related concepts, striving to provide detailed insights.

The first concept related to patience is self-control (Bettinger & Slonim, 2007; Hanks & Stratton, 2007; Morinis, 2007). Baumeister et al. (2007) describe self-control as the regulation of one's emotions, behaviors, and attention in situations involving conflicting options, and is evaluated as the ability to weigh these options, choose one, and delay the others while pursuing it. Patience in the context of self-control is considered the ability to endure uncertainty and postpone action (Duckworth et al., 2019). Studies have shown that patience and self-control are separate psychological constructs and therefore the two concepts should be measured with different scales (Khormaei et al., 2017; Schnitker et al., 2017). In a study of adolescents' patience and self-control, Khormaei et al. (2017) concluded that they are related but different psychological concepts. Consequently, while patience and self-control are interconnected, they are distinct psychological constructs that necessitate separate measurement scales for accurate assessment.

The next concept, resilience, as broadly defined, is the successful adaptation to challenging conditions (Norman, 2000). Grasping the concept of resilience involves

recognizing two pivotal components: (a) encountering difficulties or potential threats, and (b) the capability to make a positive transition in the wake of these challenges (Luthar et al., 2000; Masten, 1994; Masten et al., 1990). Patience and resilience differ in their focus and outcomes. Patience involves dealing with challenges calmly and accepting the present moment while recognizing that improvement takes time (Hanks & Stratton, 2007; Morinis, 2007). Resilience, on the other hand, entails regaining strength after difficult experiences, learning from adversity, and rising above challenging circumstances (Luthar et al., 2000; Masten, 1994). Thus, while resilience is an active response, actively overcoming obstacles for personal growth, patience is more passive, involving enduring and tolerating.

Another concept, engagement, dealt with within the contexts of pro-social institutions, schools, and classes, is defined on a class level as a "constructive, enthusiastic, willing, emotionally positive, and cognitively focused participation with learning activities in the classroom," with three essential indicators, namely behaviorist (action initiation, working hard, involvement, etc.), emotional (enthusiasm, interest, enjoyment, zest, wonder, etc.), and cognitive (follow through, strategy search, purposeful, etc.) (Skinner & Pitzer, 2012, p. 22). In other words, engagement refers to actively participating, involving oneself, or being fully present in a particular activity, task, or relationship. It involves enthusiasm, interest, and commitment. While patience emphasizes waiting and endurance, engagement focuses on active participation and being fully present in the current situation.

The last concept is persistence. In psychology, persistence is addressed in two contexts. The first context is related to school attendance. Student persistence refers to a student's commitment to completing a course or program of study. Persistence is considered a positive measure of success as opposed to dropout, which is a negative outcome (Cookson, 1988). The second context of persistence is as one of the character strengths. According to Peterson and Seligman (2004), persistence is the voluntary continuation of goal-directed action despite obstacles, difficulties, or discouragement. Simply measuring how long someone works at a task does not adequately capture the essence of perseverance because continuing to perform something fun or rewarding does not require one to endure and overcome setbacks. The main distinction is that patience relates to waiting and endurance, while persistence involves taking continuous action toward achieving a goal. In other words, persistence is about continuously exerting effort despite obstacles, whereas patience focuses on maintaining calm and steadiness during delays or challenges.

1.3 Online learning patience

Patience has been studied in the context of education. Studies have shown that as students' ability to postpone pleasure and satisfaction increases, their academic success also increases (Flynn, 1985; Shoda et al., 1990). Galla and Duckworth (2015) asserted that patience has a significant impact on students' study habits, completing homework, and attending school. Impatience on the other hand has been linked with children's and adolescents' substance addiction, academic failure, and behavioral disorders (Castillo et al., 2011; Sutter et al., 2013). Given the challenges students are likely to experience in online learning, patience is essential because learning under

any circumstances requires an investment of time and effort, and in online learning, they must cope with distraction, boredom, disengagement, lack of motivation, increased responsibility, and lack of social interaction. Based on an analysis of the relevant literature, McLoughlin and Lee (2010) concluded that in an online environment, the learner has more responsibility to "prepare for his/her own learning, take the necessary steps to learn, manage and evaluate the learning and provide self-feedback and judgement, while simultaneously maintaining a high level of motivation" (p. 29). Patience may help students regulate their learning and make them become successful in online learning environment.

In the MacMillan Dictionary (2006), patience is defined as "the ability to continue doing something for a long time without losing interest, especially something difficult" (p. 1039), and in the Longman Dictionary of Contemporary English (2004), it is defined as "the ability to continue waiting or doing something for a long time without becoming angry or anxious" (p. 1206). Despite the clarity of the presented definitions, these definitions are not directly related with learning and online learning. Within the aim of this study, we define online learning patience as *the ability to steadily continue studying or practicing in the online learning environments, even when it becomes difficult or takes a long time* (Fig. 1). It involves staying calm and not becoming upset or bored during the learning process. In other words, when faced with tough or lengthy learning tasks, learning patience is about keeping going without getting angry, anxious, or losing interest. This approach acknowledges that learning can be a slow process and that staying patient is key to successfully absorbing and mastering new information or skills. In learning patience, the focus is on fostering a sense of tranquility and positivity in situations that necessitate waiting or enduring hardships, rather than on the direct pursuit of goals (as in perseverance) or the restraint of impulses (as in self-control).

This study aims to develop and validate an Online Learning Patience Scale, addressing a significant gap in current educational research. There exist studies investigating patience, particularly in relation to teachers. Studies by Meriç and Erdem (2022) and Zhang (2022) highlight the importance of teacher patience in student academic achievement. Teachers with higher levels of patience tend to exhibit expertise in intervention strategies, effective collaboration with interdisciplinary teams, and maintain a positive approach towards students, as noted by Sherman et al. (2008). These findings highlight the value of patience in educational settings. However, a

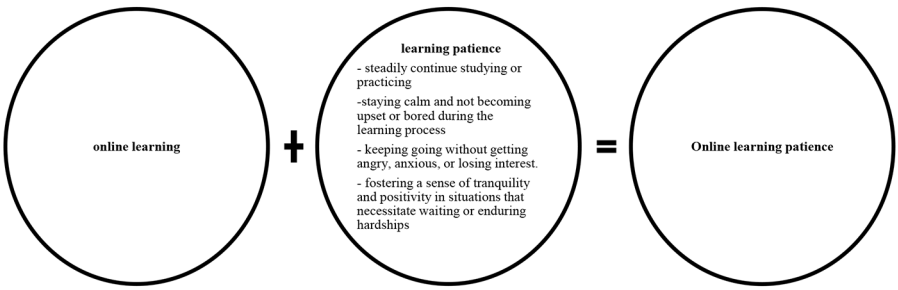


Fig. 1 Online learning patience

field-specific online learning patience scale remains absent. Patience can be evaluated in general as well as field-specifically. Based on Tooby and Cosmides' (1992) ideas, Curry et al. (2008) claim that "…there is no general-purpose ability to be patient; rather, there are numerous special-purpose, domain-specific psychological mechanisms" (p. 783). In this context, the patience scale required for learning in online environments is important.

The rationale for developing this scale is twofold. First, the concept of learning patience in online environments has not been adequately explored in existing literature. Second, the nature of learning, as described by Pace (1982), inherently demands a commitment of time and effort from the learner, mirroring the need for patience in the process of acquiring new skills and knowledge. Learning involves the use of working memory to hold information briefly before transferring it to long-term memory, a process outlined by Wickens and Carswell (2021). This process can be particularly challenging and time-consuming, stressing the need for patience, especially in online settings where learners are often required to manage their own progress because the physical contexts of online and face-to-face instruction are very different. The differences between the two contexts, and especially the physical remoteness of the teacher in online instruction, call for the measurement of learning patience in online education.

2 Method

The present study aimed to develop a scale for measuring students' patience in online learning environments. To this end, a cross-sectional research design was employed, and the scale was developed in the following stages: preparation of items for the trial form of the scale, item analysis (27% lower-upper group), exploratory factor analysis, reliability analysis based on internal consistency (Cronbach Alpha), and establishment of concurrent validity. Data were collected online on a voluntary basis from secondary and high school students continuing their education online in the 2020–2021 academic year at separate times based on the stages of our research (Table 1).

Criterion sampling is a sampling method that entails carefully selecting individuals, events, objects, or situations with specific characteristics relevant to the research problem at hand (Palinkas et al., 2015). In this specific research, the scholars employed the Risk Map of the Istanbul Provincial National Education for the 2020–2021 academic year to pinpoint schools categorized as having "moderate psychological risk." From this particular province and the identified schools, participants for the study were chosen randomly.

Table 1 Data collection stages and number of participants

Stage	Group	Participants
First	Item analysis / Exploratory Study Group (EFA)	250
Second	Confirmatory Study Group (CFA)	266
Third	Validity Study Group for Impulsivity and Test-anxiety (VIT)	486
Fourth	Validity Study Group for Self-control (VSC)	232

As a criterion for inclusion in the study, participants being in the middle adolescence period has been taken as the basis. Generally, the adolescent period is considered in 3 stages: early adolescence, middle adolescence, and late adolescence (Steinberg, 2020). The 7th, 8th, 9th, and 10th-grade students participating in the study meet the criterion of being in the middle adolescence period. The only exclusion criterion was having had a psychiatric diagnosis, determined by responses to the question of "Do you have any psychiatric disorders" on the demographic information form administered before the data were collected.

In this study, the criteria expressed in the literature were utilized for sample size. Firstly, the sample size in scale development studies can be calculated by multiplying the item number by ten (sample size = item number x 10 individuals) (Nunnally, 1978). According to this criterion, in a 10-item online learning patience scale, at least 100 people should be included in the sample. In the current study, close to 200 participants have been included in the sample for each study separate groups (4 groups): First, we conducted the Exploratory Study Group (EFA), involving 250 participants. Second, we conducted the Confirmatory Study Group (CFA) with 266 participants, aiming to validate the findings from the EFA. Following this, we undertook the Validity Study Group for Impulsivity and Test-anxiety (VIT), the largest group with 486 participants, to assess the reliability of our measures in these specific areas. Lastly, we conducted the Validity Study Group for Self-control (VSC), consisting of 232 participants.

Secondly, Guadagnoli and Velicer (1988) and Comrey (1988) state that samples of 200–300 people are sufficient for factor analysis in scale development studies. When evaluated together, it is understood that the number of participants in the current study is quite sufficient.

No ethical approval was required for this study since it is a non-invasive study. However, ethical considerations such as confidentiality of the data, anonymity of participants, informed consent, and right to withdraw from the study were guaranteed. The validity of the instruments' scales for impulsivity, self-control, and test anxiety scales was examined. In the literature, learning patience is shown to be related to psychological structures such as impulsivity and self-control. However, these relationships were examined with general patience scales and not associated with online learning patience (Caci et al., 1998; Khormaei et al., 2017; Lecrubier et al., 1995).

2.1 Preparation of the items

In the preparation of the items, the literature related to patience and patience in the learning process was first examined (Baumeister et al., 2007; Bettinger & Slonim, 2007; Flynn, 1985; Galla & Duckworth, 2015; Hanks & Stratton, 2007; Morinis, 2007; Schnitker & Emmons, 2007; Schnitker et al., 2017; Stevens et al., 2005). Next, 30 male and 30 female high school students were interviewed for determining the items in the scale. For example, open-ended questions were asked, such as "Imagine that you are losing patience in an online class. What do you think and feel at these times?" and "Think about times when your patience increased in online classes. What did you think and feel during these times" were asked? Based on the analysis of the responses, 14 scale items were determined and examined by the researchers and

experts in measurement and evaluation and educational psychology in terms of form, expression, and participation indicators. It was suggested that two items in the scale should be corrected in terms of language and meaning and 3 items should be removed from the scale item pool. As a result, the trial form of the scale had 11 items with 5 point Likert scale and was then tested in further studies.

2.2 Study groups

This study was conducted with four separate groups: Exploratory study group (EFA) with 250 participants, Confirmatory study group (CFA) with 266 participants, Validity study group for Impulsivity and Test-anxiety (VIT) with 486 participants, and Validity study group for Self-control (VSC) with 232 participants. The distribution of participants among the study groups is described below:

Exploratory study group The group in which the exploratory factor analysis was carried out included 83 boys and 167 girls, a total of 250 public secondary (junior high) school and high school) students, of whom 109 (43.6%) were seventh-grade students and 141 (56.4%) were eighth-grade students. Regarding parental education, the mothers of 71 (28.4%) had completed primary school, 56 (22.4%) secondary (junior high) school, 46 (18.4%) high school, and 77 (30.8%) undergraduate education. The fathers of 51 (20.4%) completed primary school, 74 (29.6%) secondary school, 50 (20%) high school, and 75 (30.0%) undergraduate education.

Confirmatory study group A total of 266 students, 96 (36%) were boys and 170 girls, were included in the group in which the confirmatory factor analysis was carried out, of whom 139 (52.3%) were in the seventh grade and 127 (47.7%) in the eighth-grade students. The mothers of 55 (20.7%) had completed primary school, 43 (16.2%) secondary school, 91 (18.0%) high school, and 77 (28.9%) undergraduate education. The fathers of 48 (18.0%) had completed primary school, 45 (16.9%) secondary school, 81 (30.5%) high school, and 92 (34.6%) undergraduate education.

After the initial exploratory and confirmatory factor analyses, two further analyses were conducted to differentiate the concept of ‘online learning patience’ from text anxiety and ‘impulsivity’ and ‘self-control’.

Validity study group for Impulsivity and Test anxiety: The validity study for Impulsivity and Test-anxiety was carried out with 486 secondary (junior high school) students, 167 (34%) males and 319 (66%) females, of whom 232 (47.7%) were 7th-grade students and 254 (52.3%) were 8th-grade students.

Validity study group for Self-control: This validity group had a total of 232 students, 184 (79%) girls and 48 (21%) boys, of whom 91 (39.2%) were 9th-grade students and 141 (60.8%) were 10th-grade students.

2.3 Measuring tools

As mentioned, the validity of the SOLPS was examined using impulsivity, self-control, and test anxiety scales. The psychometric features of these related scales are described below:

2.3.1 Self-control and self-management scale

This scale was developed by Mezo (2009) and adapted into Turkish by Ercoşkun (2016). The scale consists of 16 items in three sub-dimensions: "Self-Reinforcement", "Self-Assessment" and "Self-Adjustment". Items are rated with a 6-point Likert-type scale from 1) Doesn't describe me at all to 6) Describes me completely.

2.3.2 Barratt impulsivity scale short form

The first form of this scale, which consists of three subscales: (a) Inability to plan (inability to control, intolerance of cognitive confusion), (b) Motor impulsivity (impatience), and (c) Attention impulsivity (inattention and cognitive dysregulation) was developed by Barratt (1959) for individuals between the ages of 18-51and subsequently revised many times. The short form of the scale consisting of 15 items was developed by Spinella (2007). Tamam et al. (2013) adapted the scale into Turkish, and it has been used with different age groups in Turkey. Items on the BIS-11-SF are rated on a four-point Likert scale. The internal consistency of the scale calculated with Cronbach's alpha, was found to be between 0.82 for the total score and between 0.64 and 0.80 for the subscales.

2.3.3 Westside test anxiety inventory

This scale, originally developed by Driscoll (2007), was adapted into Turkish by Totan and Yavuz (2009) for university students and by Totan (2018) for high school students. The scale, in which maladaptation and anxiety constitute the only factor, consists of 11 items rated on a 5-point Likert-type scale from (1) Never True to (5) Always True. The lowest score that can be obtained from the scale, which does not include reverse-coded items, is 11, and the highest score is 55. The Cronbach alpha internal consistency coefficient was found to be 0.87 for university students and 0.91 for high school students on the scale on which general test anxiety level was determined. Levels were grouped in the following ranges: Low-level test anxiety: 11.0–25.0 point range, Medium-level test anxiety: 26.0–40.0 point range, High-level test anxiety: 41.0–55.0 point range. For this study, the Cronbach alpha coefficient was calculated as 0.85, while confirmatory factor analysis (CFA) analysis indicated that all items of the scale were significantly valid, and the fit indexes were good (X2:79.3, df:44, X2/df:1.8, RMSEA: 0.034).

3 Findings

3.1 Item analysis

Before the *exploratory factor analysis* was conducted, an item analysis was conducted on the lower and upper 27% *exploratory* group ($n=250$) (Table 2) in which the independent groups t-test technique was used. In addition, item-total score correlations had values between 0.68 and 0.87. According to the results of the analysis, it was concluded that the discrimination levels of the items were high and all could be included in the factor analysis (Field, 2013).

3.2 Factor structure of the scale

Results exploratory factor analysis To examine the factor structure of the Student Patience in Online Courses Scale, factor analysis was performed on the data obtained from high school students based on Principal Component Analysis (PCA). At this point, the Kaiser-Meyer-Olkin (KMO) coefficient and the Barlett test values were also examined. The KMO value was 0.952 ($p<0.01$), and the Barlett`s Test of Sphericity Chi-Square value was 2111.386 ($p<0.01$) The KMO value showed that the sample size was sufficient, and the Barlett test results showed that the sample met the multivariate normality assumption.

Table 2 Results of item analysis

Item	Group	*n*	Mean	Std. D	t
1.	Lower (27%)	67	1.19	0.43	-12.552**
	Upper (27%)	67	3.42	1.38	
2.	Lower (27%)	67	1.34	0.84	-11.022**
	Upper (27%)	67	3.19	1.08	
3.	Lower (27%)	67	1.18	0.42	-14.102**
	Upper (27%)	67	3.29	1.15	
4.	Lower (27%)	67	1.28	0.69	-16.105**
	Upper (27%)	67	3.83	1.09	
5.	Lower (27%)	67	1.09	0.33	-14.419**
	Upper (27%)	67	3.22	1.16	
6.	Lower (27%)	67	1.12	0.32	-11.740**
	Upper (27%)	67	3.14	1.37	
7.	Lower (27%)	67	1.45	0.58	-15.856**
	Upper (27%)	67	3.83	1.08	
8.	Lower (27%)	67	1.28	0.54	-14.069**
	Upper (27%)	67	3.54	1.19	
9.	Lower (27%)	67	1.13	0.34	-19.812**
	Upper (27%)	67	3.54	0.93	
10.	Lower (27%)	67	1.42	0.63	-20.439**
	Upper (27%)	67	4.04	0.84	
11.	Lower (27%)	67	1.70	0.871	-14.223**
	Upper (27%)	67	4.03	1.02	

** $p<0.01$; * $p<0.05$

Based on the assumption that the factors in the study could be dependent and related to each other, the direct oblique rotation technique, one of the rotation techniques, was used in the exploratory factor analysis. In this analysis, starting with one factor entered and continuing to the total of 11 items, there were no items with factor loading values below 0.30. Based on the results of the exploratory factor analysis, a one-dimensional scale with a total variance of 67.80% was obtained (Fig. 2). According to the results of the analysis, the factor loading values of the items included in the scale ranged between 0.69 and 0.89 (Table 3).

3.2.1 Confirmatory factor analysis results

Confirmatory factor analysis (CFA) was used to determine the construct validity of the Student Patience in Online Courses Scale. One item with a low factor load value had been removed from the scale. According to the results of the analysis, the factor structure of the single-factor, a 10-item scale was confirmed (Table 3). The CFA fit index values were as follows: x2=87.04, sd=35, RMSEA=0.075, NFI=0.97, NNFI=0.98, CFI=0.98, IFI=0.98, RFI=0.96, IFI=0.96, GFI=0.94, AGFI=0.90 (Fig. 3). According to Kline (2015), fit index values can be acceptable when higher than 0.90, and the factor structure of the scale is confirmed when the RMSEA value is lower than 0.80.

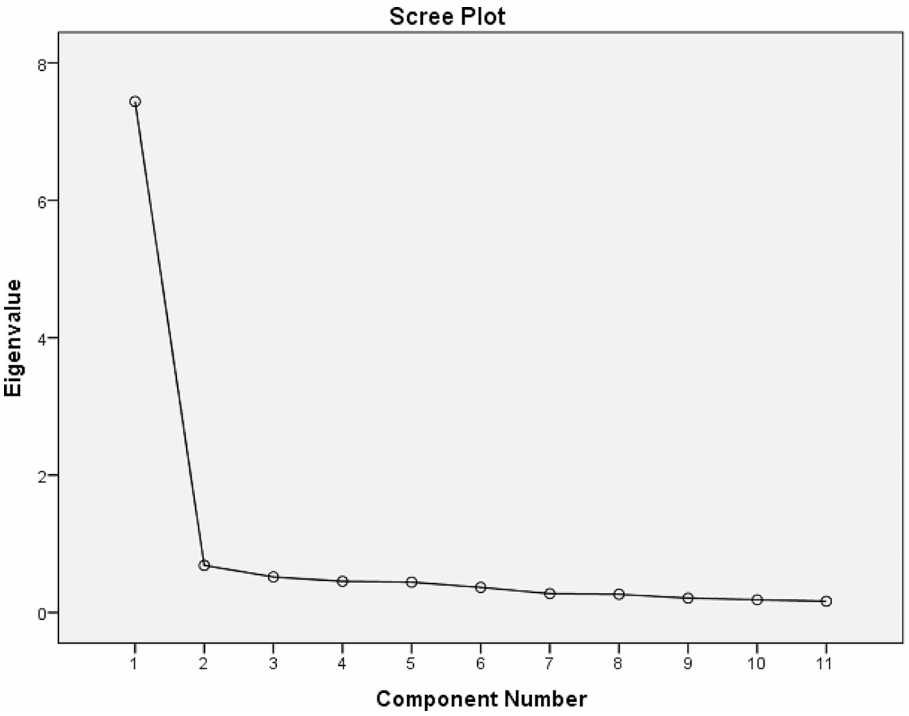


Fig. 2 Scatter diagram

Table 3 Results of the exploratory factor analysis (One-dimensional-factor scale)

Item No.	Items	Factor Loading
1	During the online course, I become impatient and find myself dealing with the internet, cell phone, and tablet, which can hinder my learning.	0.886
2	I cannot be patient when learning subjects although the course materials are well-prepared for my online courses.	0.876
3	I cannot continuously listen to the lectures in my online courses.	0.874
4	Although the subjects in my online courses are within my comprehension abilities, I get bored.	0.867
5	Although the courses I take in my online education are suitable for me, I cannot be patient while learning them.	0.864
6	When I study for my online courses, I am impatient and soon stop studying.	0.855
7	I look forward to my lecturer finishing the online class.	0.798
8	When I review my course subjects outside of the lesson, I become restless and stop studying.	0.796
9	I cannot pay attention to the subjects in my online courses for long.	0.767
10	In online courses, my mind shifts away from the course subject to unrelated subjects.	0.740
11	The time I devote to the homework given in the online courses I take seems very long to me.	0.698

The reliability of Student Patience in Online Courses Scale was carried out on the confirmatory factor analysis group. The reliability of the scale was analyzed with the Cronbach Alpha internal consistency technique, which resulted in a value of 0.95, indicating that the scale has high reliability.

3.3 Concurrent validity of the student online learning patience scale

Table 4 shows that the Student Online Learning Patience Scale is moderately and positively correlated with attentional impulsiveness ($r=0.555$; $p<0.01$) and test anxiety ($r=0.505$; $p<0.01$). On the other hand, it is seen that the scale is weakly and positively related to Non-planning Impulsiveness ($r=0.284$; $p<0.01$) and Motor Impulsiveness ($r=0.399$; $p<0.01$). These results show that the Student Online Learning Patience Scale has a different structure from other structures.

Table 5 shows that the Student Online Learning Patience Scale is moderately and negatively correlated with the total score of self-control and self-management (r=-0.444; $p<0.01$) and self-monitoring (r=-0.456; $p<0.01$). On the other hand, it is seen that the Student Online Learning Patience Scale is weakly and negatively related to self-evaluating (r=-0.306; $p<0.01$) and self-reinforcing (r=-0.274; $p<0.01$). It is also seen that the Student Online Learning Patience Scale is highly related to test anxiety (r=-0.505; $p<0.01$).

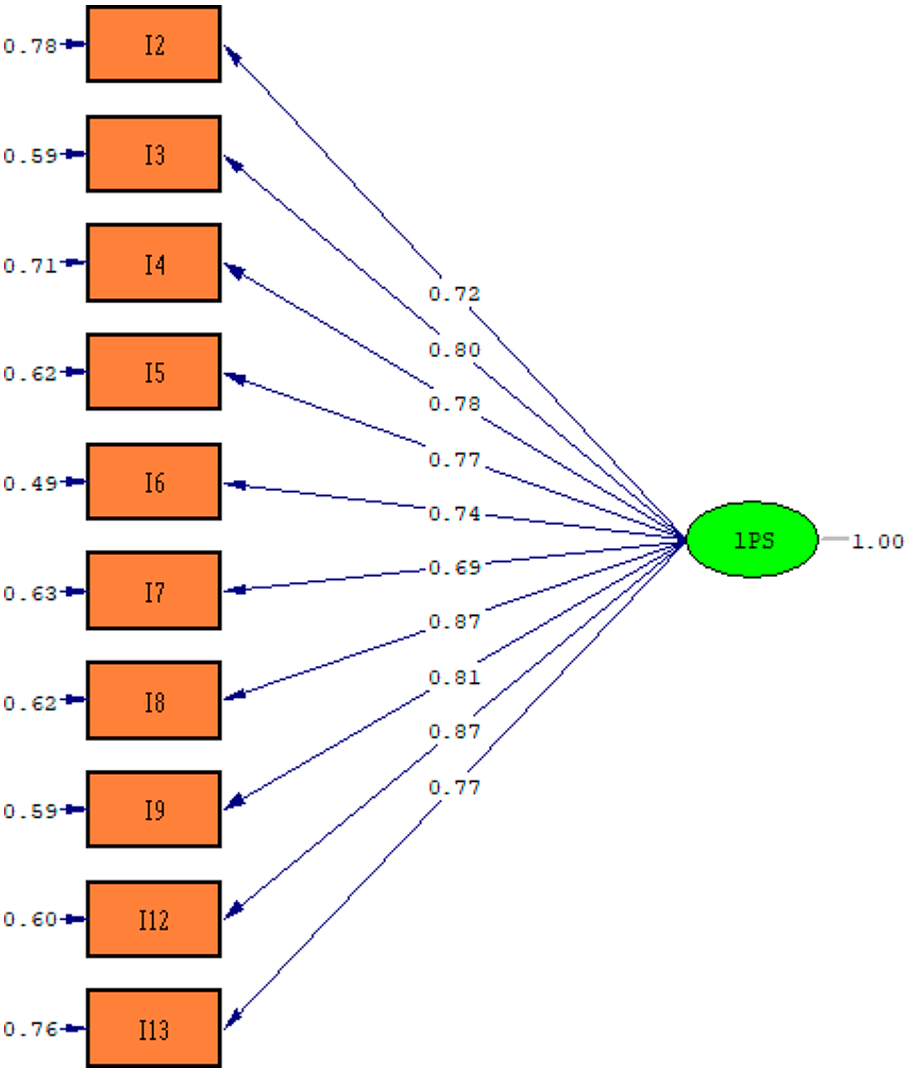


Fig. 3 Results of confirmatory factor analysis

Table 4 The Relationship of the Student Online Learning Patience Scale with Impulsivity and Test Anxiety

	Variables				
	1	2	3	4	5
1. Online patience	1				
2. Non-planning Impulsiveness	0.284**	1			
3. Attentional Impulsiveness	0.515**	0.375**	1		
4. Motor Impulsiveness	0.399**	0.233**	0.526**	1	
5. Test Anxiety	0.505**	0.225**	0.391**	0.273**	1

Note: ** $p<0.01$; * $p<0.05$

Table 5 The relationship between the Student Online Learning Patience Scale and Self Control

	Variables				
	1	2	3	4	5
1. Online learning patience	1				
2. Total score self-control and self-management	−0.444**	1			
3. Self-monitoring	−0.456**	0.868**	1		
4. Self-evaluating	−0.306**	0.545**	0.362**	1	
5. Self-reinforcing	−0.274**	0.842**	0.554**	0.210**	1

Note: ** $p<0.01$; * $p<0.05$

These results show that the Student Online Learning Patience Scale has a different structure from other scales.

4 Discussion

In this study, a scale to measure online learning patience was developed. The factor structure, reliability, and validity of the scale were tested. According to the results, the student online learning scale is a valid and reliable measurement tool.

It is important for people to have patience, which is considered a strength of character. Studies have shown that patient individuals are more cooperative (Curry et al., 2008; Stevens & Hauser, 2004). Patience is necessary for the healthy functioning of a family and the prevention of child abuse and neglect (Hanks & Stratton, 2007), and it is also effective for children's healthy consumption and money management habits (Bettinger & Slonim, 2007). Patient individuals cope better with negative situations, and as people's patience levels increase, their hope levels also rise (Khormaei et al., 2014).

When the studies on patience are examined in the literature, it is seen that, first, patience has been investigated as a general rather than a field-specific attribute (Curry et al., 2008). In some studies, patience is examined in relation to impulsivity (Baumeister et al., 2007; Bettinger & Slonim, 2007; Curry et al., 2008; Flynn, 1985; Galla & Duckworth, 2015; Hanks & Stratton, 2007; Morinis, 2007; Schnitker & Emmons, 2007; Schnitker et al., 2017; Stevens et al., 2005). Prior to the present study, a tool to measure patience specifically in the online learning process has not been developed. Thus, this study has made an important contribution to the literature.

In this study, it was found that the levels of online learning patience are moderately related to self-control and impulsivity. In some studies, the level of patience is considered as a sub-dimension of impulsivity. This study supports online learning patience as a different concept from both impulsivity and self-control.

It should be noted that items in the SOLPS Scale are worded to reflect impatience so that a high score indicates low online learning patience-impatience, and if the items are reverse-coded a high score indicates high learning patience. Thus, the scale can be used to emphasize either impatience or patience depending on the user's purpose.

Researchers have identified many problems that may be experienced in the online learning process (Almanthari et al., 2020; Bernard & Rubalcava, 2000; Capdeferro & Romero, 2012; Chang, 2020; Downes, 2016; Muilenburg & Berge, 2005). Thus, students need many personal resources to cope with the problems they encounter, of which patience is an especially valuable characteristic that can be learned and developed (Alan & Ertac, 2018). Thus, the scale developed in this study can be used in the design of teaching strategies for helping students acquire online learning patience.

5 Conclusion

Prior to this research, patience was mainly measured as a general character trait. In this research, patience was contextualized for online learning. This study is to our knowledge the first to deliver a statistically validated and reliable scale to measure student online learning patience. We hope that the scale will contribute to research that explore ways to support students' learning patience in online education environments. The finalized SOLPS scale is presented in Appendix 1. The scale can be used to create teaching methods and strategies that increase students' online learning patience. Online instructors may use the SOLPS scale to understand and promote this important character trait in their students and make informed choices of instructional techniques to better serve their learners in an online learning environment.

Appendix: Student Online Learning Patience Scale (SOLPS)

This scale has been prepared to measure your patience in online learning processes. As you fill out the scale, evaluate each expression for its suitability for you. Please, do not skip any statements.

Items of the Student Online Learning Patience Scale	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
1. During the online course, I become impatient and find myself dealing with the internet, cell phone, and tablet, which can hinder my learning.					
2. I cannot be patient when learning subjects although the course materials are well-prepared for my online courses.					
3. I cannot continuously listen to the lectures in my online courses.					
4. Although the subjects in my online courses are within my comprehension abilities, I get bored.					
5. Although the courses I take in my online education are suitable for me, I cannot be patient while learning them.					

Items of the Student Online Learning Patience Scale	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
6. When I study for my online courses, I become impatient and soon stop studying.					
7. I look forward to my lecturer finishing the online class.					
8. When I review my course subjects outside of the lesson, I become restless and stop studying.					
9. I cannot pay attention to the subjects in my online courses for long.					
10. In online courses, my mind shifts away from the course subject to unrelated subjects.					

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Data availability The datasets used and analyzed during the current research are available from the corresponding author on reasonable request.

Declarations

Conflict of interest No potential conflict of interest was reported by the author(s).

Informed consent Informed consent was obtained from all individual participants included in the study.

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